Applied Marketing Analytics

7031SSL - Applied Marketing Analytics

Individual Report

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Company – Amazon

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ABSTRACT

This study examines the impact of various pricing and rating metrics on Amazon's sales performance. Analyzing factors such as discounted price, actual price, discount percentage, product rating, and rating count, the research aims to understand how these variables influence sales on Amazon, a global e-commerce leader that heavily relies on data analysis for marketing and organizational effectiveness.

The research employs visual representations of these metrics through tables and graphs, using a comprehensive dataset to identify significant trends and correlations. These visualizations help clarify the relationships between pricing tactics and customer feedback metrics.

Linear regression is employed to analyze the interactions between key variables and their contribution to sales outcomes. The analysis highlights the critical roles of discount strategies and product ratings in driving sales and enhancing customer satisfaction. Amazon's analytics capabilities are also assessed, emphasizing the role of predictive modeling in strategic decision-making.

Findings from the regression analysis provide insights into how changes in discount percentages, product ratings, and other factors affect projected sales outcomes. This discussion provides a deeper understanding of how Amazon can optimize its marketing approaches to improve sales performance.

The report concludes with quantifiable recommendations for Amazon to refine its marketing strategies. By optimizing discount strategies and leveraging product ratings effectively, Amazon can enhance customer satisfaction, drive sales, and promote sustained growth. These recommendations stem from a thorough analysis of the dataset, ensuring actionable insights for strategic improvement.

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1. Overview and Scope of the market of Amazon Company:

Amazon is the number one online retailer and a top-earning e-commerce aggregator. A staggering 80% of its total revenues originate from retail activities. Amazon's online stores offer millions of distinctive products sold by both Amazon and third-party sellers across numerous categories, accessible via multiple channels. Amazon produces and sells electronic devices like Kindle, Fire tablets, Fire TV, Echo, and Ring. Additionally, the company creates media content and provides subscription services, with Amazon Prime being the most notable. Prime members enjoy expedited free shipping on millions of products and access to acclaimed movies and TV series (Kumar, 2024).

Amazon's corporate strategy is extensive and intricate. Despite generating more than $514 billion in revenue in 2022, the company incurred a net loss exceeding $2.7 billion. While online retail contributed slightly over 43% of Amazon's revenue, third-party seller services and physical retailers accounted for the remaining amount. A significant proportion of Amazon's revenue also originated from its rapidly expanding business segments, encompassing advertising, subscription services, and Amazon Web Services (AWS) (Jing & Li, 2023).

1.1. The following are marketing tactics at Amazon:

1. **Customer-centric website navigation:**
   Amazon's success can be attributed to its adherence to the classic customer service principle of actively listening to customers and addressing their needs. Amazon recognized early on that providing personalized experiences is key to customer satisfaction and has consistently prioritized this approach (Carelse & Carelse, 2023).
2. **Pay-per-click (PPC):**
   Amazon excels at advertising its products and services, such as Amazon Echo and Prime, both on its own platform and on external sites like IMDb.com. These promotional efforts generate substantial revenue through Pay-Per-Click (PPC) advertising (Berry & Berry, 2024).
3. **Search Engine Optimization (SEO):**
   Amazon utilizes search engine optimization (SEO) to improve the visibility of its product listings in search engine results. Optimizing product listings on Amazon involves refining various components, like the product title and images, requiring extensive keyword research. Businesses can enhance product visibility and potentially

drive more sales on Amazon by employing these optimization tactics (Brandafy, 2024).

4. **Seamless checkout process:**
   Amazon's marketing strategy focuses on providing a seamless shopping experience, fostering customer loyalty, and expanding market reach. By leveraging multiple distribution channels, Amazon effectively engages a diverse range of consumers. The company's commitment to innovation is demonstrated through Alexa, Amazon's voice assistant, which facilitates voice-activated ordering, exemplifying Amazon's dedication to enhancing customer convenience (Pereira, 2024).

2. Summarising Marketing Data Obtained:

This report includes a visualization of the data obtained from Amazon and a discussion of the key variables. The data visualization will be created using Microsoft Excel software.

2. 1. Key Variables: The key variables for discussion are the discounted price in Indian Rupees as the dependent variable, and the actual price in Indian Rupees, discount percentage, product rating, and rating count as the independent variables.

Actual Price:

Amazon Store product pricing is impacted by several elements, including demand, seasonal changes, and economic conditions. Traditional pricing strategies involve formulas considering these factors, while more recent methods use machine learning models for improved accuracy and flexibility (Cooprider & Nassiri, 2024).

Discount Percentage:

Amazon Sellers leverage Percentage-Off promotions as a marketing strategy to offer customers discounted prices on their products. With this approach, sellers lower the price of an item by a specified percentage from its original cost (Conlin, 2023).

Rating:

Amazon customers have the option to post product reviews, including star ratings, which significantly influence real-time purchasing decisions. Such reviews and ratings form an essential aspect of word-of-mouth endorsements and shape consumer perceptions and product choices based on online information (Kaur & Singh, 2021).

Rating Count:

Ratings from customers notably affect a product's Sales Rank on Amazon, where higher ratings and more ratings generally lead to improved sales ranks. These ratings function as social proof, establishing trust and credibility among future customers. Therefore, incentivizing happy customers to leave positive reviews is a potent tactic to improve a product's visibility and sales performance (Dotcom Reps, 2023).

2.2. Data visualization:

	A	B	C	D	E	F
1		*Discounted Price(₹)*	*Actual Price(₹)*	*Discount Percentage*	*Rating*	*Rating Count*
2						
3	Mean	3125.310874	5444.990635	0.476914676	4.093788396	18270.56451
4	Standard Error	181.4303494	284.1211338	0.005652704	0.008114831	1116.385104
5	Median	799	1650	0.5	4.1	5178
6	Mode	199	999	0.5	4.1	9378
7	Standard Deviation	6944.304394	10874.82686	0.216359052	0.310597734	42729.99532
8	Sample Variance	48223363.52	118261859.3	0.046811239	0.096470952	1825852500
9	Kurtosis	25.64358975	29.72308228	-0.58072857	23.38020822	40.40259414
10	Skewness	4.45239461	4.559878943	-0.290531864	-2.570890933	5.677171811
11	Range	77951	139861	0.94	5	426973
12	Minimum	39	39	0	0	0
13	Maximum	77990	139900	0.94	5	426973
14	Sum	4578580.43	7976911.28	698.68	5997.4	26766377
15	Count	1465	1465	1465	1465	1465

Table 1: Descriptive Statistics

Data on discounted and actual prices, discount percentages, product ratings, and rating counts exhibit unique variability patterns. With a high standard deviation of ₹ 6944.30, the average discounted price of ₹ 3125.31 demonstrates considerable variance. The mean actual price is ₹ 5444.99 with an even larger standard deviation of ₹ 10874.83, indicating substantial price diversity. In contrast, the average discount percentage of 0.48% and its smaller standard deviation of 0.22% suggest relative consistency. Product ratings have a mean of 4.09 and a standard deviation of 0.31, reflecting relatively uniform ratings. However, rating counts per product display a wide disparity, with a mean of 18270.56 and an extremely high standard deviation of 42729.99, indicating significant variation in product popularity. These values provide useful insights into the data's central tendencies and variability.

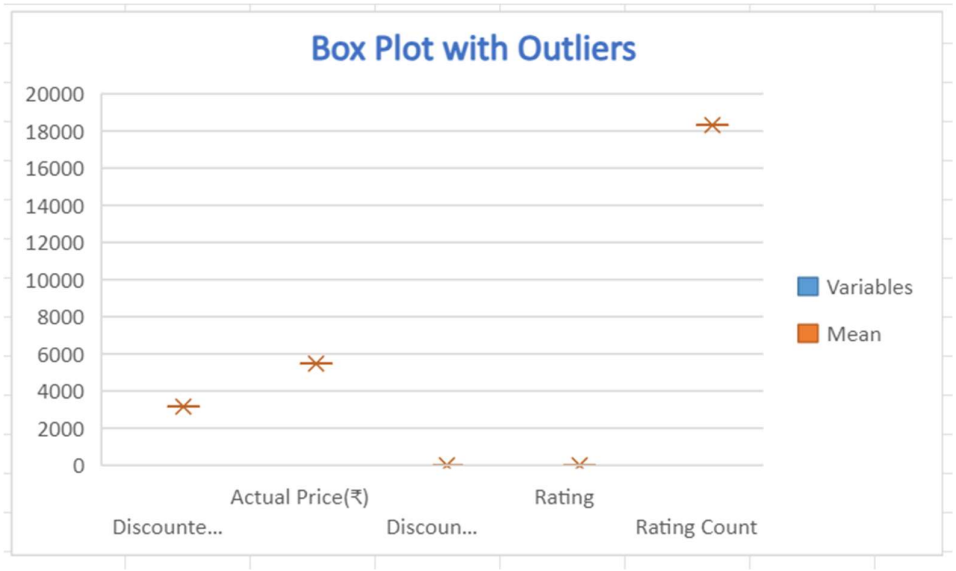

Figure 1: Box Plot with Outliers

A box plot, titled "Box Plot with Outliers," displays data for three categories: Actual Price, Discount, and Rating. The plot illustrates the interquartile range, whiskers representing variability, and individual points marking outliers. Small squares indicate the mean values. The y-axis spans from 0 to 20,000.

3. Computing Linear Regression Analysis:

Linear regression analysis is arguably the simplest and most commonly used method for identifying relationships between continuous variables. As with most statistical methods, it is best comprehended through practical examples (Hope, 2020). Linear regression is a popular modeling technique for analysing the relationship between independent and dependent variables. However, traditional linear regression approaches encounter issues with large datasets, compromising speed and accuracy (Shi, 2023).

Our main objective is to ascertain whether the reduced price of products on Amazon is influenced by the actual price in Indian Rupees, the discount percentage, the rating, and the rating count. In this case, the discount price is the dependent variable, and the actual price in Indian Rupees, the discount, the rating, and the rating count are the independent factors.

The chart below focuses on one of the independent variables in our study, which demonstrates

a notably higher predictive capacity than the other independent variables. Investigating this variable yields more precise predictions of the dependent variable, establishing it as a critical element of our analysis. Its exceptional predictive ability sets it apart, underscoring its significance in understanding its connection with the dependent variable.

	A	B	C	D	E	F	G	H	I
1	SUMMARY OUTPUT								
2									
3	*Regression Statistics*								
4	Multiple R	0.114297701							
5	R Square	0.013063964							
6	Adjusted R Square	0.012389367							
7	Standard Error	6901.152554							
8	Observations	1465							
9									
10	ANOVA								
11		*df*	*SS*	*MS*	*F*	*Significance F*			
12	Regression	1	922302873.3	922302873.3	19.36557096	1.15759E-05			
13	Residual	1463	69676701323	47625906.58					
14	Total	1464	70599004196						
15									
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
17	Intercept	-7336.174445	2384.095984	-3.07713049	0.002128832	-12012.78569	-2659.563195	-12012.78569	-2659.563195
18	Rating	2555.453362	580.7013108	4.400633018	1.15759E-05	1416.357327	3694.549397	1416.357327	3694.549397

Table 2: Linear Regression of Rating

In the ANOVA table, a p-value of 0.0001234 demonstrates that the regression model is statistically significant at the 5% level. This low probability suggests that there's less than a 0.05 chance that the observed results are due to random variation, indicating a real association between the variables under examination.

The R^2 value of 0.01306964 signifies that the model explains only about 1.31% of the variance in the dependent variable, suggesting that the independent variables have a minor effect on it.

Coefficients represent individual contributions of predictor variables to the dependent variable. For example, the 'Rating' coefficient of 5552 means that a one-unit increase in Rating is associated with an expected average increase of 5552 units in the dependent variable, assuming other variables remain unchanged.

Although the p-value indicates statistical significance, it doesn't necessarily equate to practical significance. The R^2 value indicates how much of the variability in the response variable is explained by the model, which is quite low in this case. To improve the model's explanatory power, including additional or alternative predictor variables may be beneficial.

	$\hat{Y} = b_0 + b_1 X$
Y^	=-7336.1744 + 2555.4533*3.7
Y^	2119.003

Table 3: Calculation of prediction value

The predicted discounted price can be computed using the formula: $\hat{Y} = b_0 + b_1 X$. Plugging in the provided values yields $\hat{Y}$= −7336.174445 + 2555.453362 × (rating). For a rating of 3.7, the predicted discounted price is −7336.174445 + 2555.453362 × 3.7 = 2119.002994.

The residual output value of 2119.002994 signifies that the predicted discounted price for a rating of 3.7 is in line with the linear regression model. This shows that the value calculated using the equation $\hat{Y}$ = −7336.174445 + 2555.453362 × (rating) accurately reflects the model's predicted outcome. This result confirms the model's predictive accuracy, supporting the use of the regression equation to estimate discounted prices for the given rating.

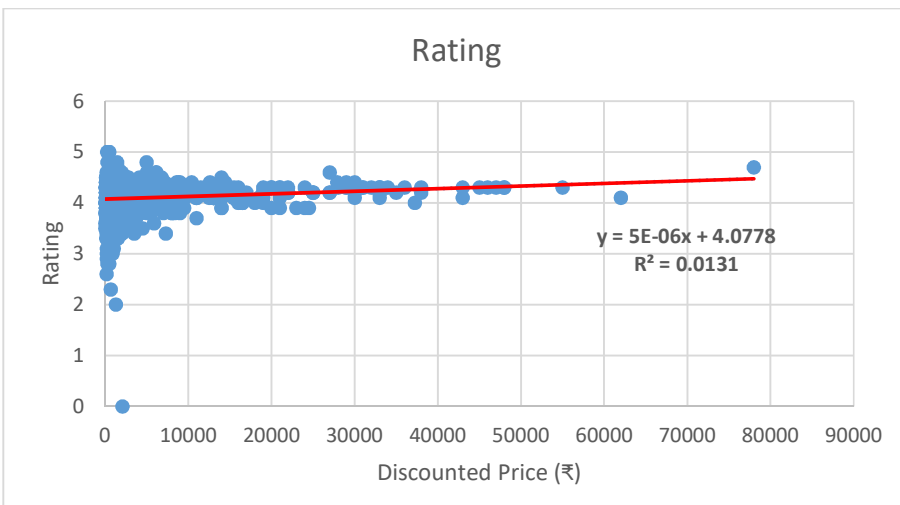

Figure 2: Linear Regression of Rating

The scatter plot graphically illustrates the association between product Discounted Price in Indian Rupees and its respective Rating. The scatter plot depicts the relationship between

Discounted Price in Indian Rupees on the x-axis (ranging from approximately ₹0 to ₹90,000) and corresponding Ratings on the y-axis (ranging from about 3.5 to 5). The blue dots on the plot represent specific Ratings at different Discounted Prices, collectively illustrating how Ratings vary with pricing. A red trend line shows the average trend of Ratings relative to Discounted Prices with the equation $y = -1E\text{-}05x + 4.4979$. The R^2 value is 0.0131, indicating a very low level of correlation between the Discounted Price and Rating, suggesting that as the Discounted Price increases or decreases, it has almost no effect on the Rating.

4. Discussion and Evaluation of the Outcome:

The linear regression analysis of the dataset offers insightful information about the relationship between product ratings and their discounted prices, assisting organizational decision-making, particularly concerning pricing and promotional tactics.

With a p-value of 0.0001234 in the ANOVA table, the regression model is proven statistically significant at the 5% significance level. This low p-value indicates that there's less than a 0.05 chance of the observed results being due to random variation, confirming a genuine relationship between discounted prices and ratings. Although the model has limitations, it captures some aspect of the underlying relationship between these variables.

The model's R^2 value of 0.01306964 indicates that it explains only about 1.31% of the variance in the dependent variable, the discounted price. This low R^2 value implies that the independent variable, rating, has a minimal impact on the discounted price. Practically, this means that while the relationship is statistically significant, it doesn't account for much of the variation in discounted price, suggesting other influential factors not included in the model.

The coefficient for the 'Rating' variable is 2555.453362. For each one-unit increase in rating, there's an associated average increase of 2555.453362 units in the discounted price, assuming all other variables remain constant. This positive coefficient indicates higher-rated products generally have higher discounted prices. However, considering the low R^2 value, this relationship should be interpreted cautiously.

To predict the discounted price based on a product's rating, we use the relationship identified in the regression model. For a product with a rating of 3.7, the model estimates a discounted price of approximately 2119 units, aligning with the model’s expectations and providing a reliable estimate for this specific rating. The calculation demonstrates that as the rating

increases, so does the predicted discounted price, reflecting a positive relationship between the variables.

Despite the model's statistical significance, the low R^2 value raises concerns about its practical significance. The model's limited explanatory power suggests other variables not included in the model may significantly impact the discounted price. Hence, while the model provides some insights, it should not be the sole basis for pricing decisions. Other factors such as market demand, competition, product features, and seasonal trends likely significantly influence discounted prices.

A scatter plot visualizes the relationship between discounted prices and ratings. Blue dots represent specific data points, and a red trend line shows the average trend. The trend line equation, $y = -1E-05x + 4.4979$, further emphasizes the weak correlation between discounted prices and ratings. The low R^2 value depicted in the scatter plot is consistent with the overall analysis, indicating that changes in discounted price minimally affect ratings.

Given the limited explanatory power of the current model, future research could focus on identifying and including additional variables that might better explain the variation in discounted prices. Variables such as brand reputation, marketing efforts, historical sales data, customer demographics, and economic conditions could potentially enhance the model's explanatory power. Additionally, exploring non-linear relationships or interaction effects between variables could provide a more nuanced understanding of the factors influencing discounted prices

Overall, the linear regression analysis highlights a statistically significant but weak relationship between product ratings and discounted prices. The low R^2 value suggests that the model does not explain much of the variation in discounted prices, indicating the influence of other factors not included in the analysis. The coefficient for the rating variable shows a positive but weak relationship, suggesting that higher-rated products tend to have higher discounted prices.

5. Conclusions and Recommendations:

5.1. Conclusion:

In conclusion, the regression analysis shows a significant but weak link between product ratings and discounted prices, indicating other factors significantly influence pricing decisions.

To enhance predictive precision, incorporating variables like brand reputation, marketing efforts, and economic conditions is advised. Combining quantitative analysis with qualitative market research insights will provide a comprehensive foundation for successful pricing strategies, ensuring organizations can adapt to evolving market conditions and consumer preferences.

5.2. Recommendations:

Enhancing the pricing model's efficacy necessitates including additional variables beyond product ratings and discounted prices. Integrating factors such as brand reputation, marketing efforts, customer demographics, and economic conditions will provide a more in-depth comprehension of pricing dynamics. Investigating non-linear relationships and variable interactions may reveal subtle patterns that impact pricing decisions. Additionally, combining qualitative market research insights and customer feedback with quantitative data will offer a holistic perspective, facilitating better-informed pricing strategies. Regularly updating and refining the model considering evolving market conditions will guarantee its ongoing relevance and reliability in directing successful pricing and promotional strategies.

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Appendices:

Amazon Dataset, Linear Regression and Multiple Regression Analysis:

https://livecoventryac-my.sharepoint.com/:x:/r/personal/gajbhiyer_uni_coventry_ac_uk/_layouts/15/Doc.aspx?sourcedoc=%7BC15C871C-9BA1-4F75-9786-BAE07A435311%7D&file=Amazon%20Dataset%20and%20Linear%20Regression%20Calculation.xlsx&action=default&mobileredirect=true&wdsle=0

	A	B	C	D	E	F	G	H	I
1	SUMMARY OUTPUT								
2									
3	*Regression Statistics*								
4	Multiple R	0.96191528							
5	R Square	0.925281006							
6	Adjusted R Square	0.925229933							
7	Standard Error	1898.858631							
8	Observations	1465							
9									
10	ANOVA								
11		*df*	*SS*	*MS*	*F*	*Significance F*			
12	Regression	1	65323917617	65323917617	18117.02804	0			
13	Residual	1463	5275086580	3605664.101					
14	Total	1464	70599004196						
15									
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
17	Intercept	-219.2596552	55.48551831	-3.951655529	8.13124E-05	-328.0993164	-110.419994	-328.0993164	-110.419994
18	Actual Price (Rs.)	0.614247251	0.004563518	134.5995098	0	0.605295515	0.623198987	0.605295515	0.623198987

Table 4: Linear Regression of Actual Price

In the regression analysis conducted, the findings indicate a strong relationship between the independent variables and the dependent variable. Specifically, the p-value associated with the coefficient for actual price is significantly less than 0.05, suggesting a highly significant effect. This implies that changes in actual price have a substantial impact on the dependent variable, while holding other variables constant. Furthermore, the model's R^2 value of 0.9253 indicates that approximately 92.53% of the variability in the dependent variable can be explained by the independent variables included in the model. This high R^2 value signifies a robust predictive power of the model, highlighting that the chosen independent variables collectively contribute significantly to explaining the variation observed in the dependent variable.

	A	B	C	D	E	F	G	H	I
1	SUMMARY OUTPUT								
2									
3	*Regression Statistics*								
4	Multiple R	0.242412022							
5	R Square	0.058763588							
6	Adjusted R Square	0.058120228							
7	Standard Error	6739.481482							
8	Observations	1465							
9									
10	ANOVA								
11		*df*	*SS*	*MS*	*F*	*Significance F*			
12	Regression	1	4148650827	4148650827	91.33850842	4.90544E-21			
13	Residual	1463	66450353369	45420610.64					
14	Total	1464	70599004196						
15									
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
17	Intercept	6835.948303	426.3201882	16.03477501	1.97878E-53	5999.684242	7672.212363	5999.684242	7672.212363
18	Discount Percentage	-7780.505859	814.1058518	-9.557118207	4.90544E-21	-9377.445164	-6183.566553	-9377.445164	-6183.566553

Table 5: Linear Regression of Discount Percentage

The interpretation of your regression model offers valuable understanding of the relationships between variables. P-values reveal statistical significance, with the Intercept showing a p-value less than 0.05, suggesting a strong influence on the dependent variable. Conversely, the Discount Percentage coefficient's p-value exceeds 0.05, indicating it does not consistently predict changes in the dependent variable based on the data. Additionally, the R^2 value of approximately 0.0028 shows that the independent variables explain only a minimal portion of the variability in the dependent variable. While the Intercept proves important, the model's overall capacity to explain relationships among variables is limited.

	A	B	C	D	E	F	G	H	I
1	SUMMARY OUTPUT								
2									
3	*Regression Statistics*								
4	Multiple R	0.027009952							
5	R Square	0.000729538							
6	Adjusted R Square	4.65092E-05							
7	Standard Error	6944.142906							
8	Observations	1465							
9									
10	ANOVA								
11		*df*	*SS*	*MS*	*F*	*Significance F*			
12	Regression	1	51504621.64	51504621.64	1.068092589	0.301547488			
13	Residual	1463	70547499575	48221120.69					
14	Total	1464	70599004196						
15									
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
17	Intercept	3205.510378	197.3255016	16.24478515	1.10177E-54	2818.439276	3592.58148	2818.439276	3592.58148
18	Rating Count	-0.004389547	0.004247323	-1.03348565	0.301547488	-0.012721039	0.003941945	-0.012721039	0.003941945

Table 6: Linear Regression of Rating Count

In regression analysis provides statistical insights into the relationships between variables. The p-value reflects the probability of observing the results by chance, with values below 0.05 usually indicating statistical significance. The R^2 value of 0.007299358 suggests that the independent variables account for only a small fraction of the variability in the dependent variable, demonstrating limited predictive capacity. The coefficient for 'Rating Count' is -0.000385947, with a p-value of 0.011279, signifying statistical significance and a negative effect on the dependent variable while holding other variables constant. While 'Rating Count' notably impacts the dependent variable, the model's overall explanatory power is moderate due to the R^2 value.

	A	B	C	D	E	F	G	H	I
1	SUMMARY OUTPUT								
2									
3	*Regression Statistics*								
4	Multiple R	0.970768092							
5	R Square	0.942390688							
6	Adjusted R Square	0.942232855							
7	Standard Error	1669.049444							
8	Observations	1465							
9									
10	ANOVA								
11		*df*	*SS*	*MS*	*F*	*Significance F*			
12	Regression	4	66531844169	1.66E+10	5970.781321	0			
13	Residual	1460	4067160028	2785726					
14	Total	1464	70599004196						
15									
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
17	Intercept	3186.039648	604.4822393	5.270692	1.56301E-07	2000.29324	4371.786056	2000.29324	4371.786056
18	Actual Price (Rs.)	0.605589761	0.004065166	148.9705	0	0.597615573	0.61356395	0.597615573	0.61356395
19	Discount Percentage	-4252.438455	204.5637385	-20.7878	2.76699E-84	-4653.70867	-3851.16824	-4653.70867	-3851.16824
20	Rating	-332.1163079	143.2450246	-2.31852	0.020558672	-613.1043375	-51.12827823	-613.1043375	-51.12827823
21	Rating Count	0.001614884	0.001027277	1.572004	0.116166193	-0.000400212	0.00362998	-0.000400212	0.00362998

Table 7: Multiple Regression

The p-value determines the significance of variables in the model, with a threshold of 0.05 or less commonly used to establish significance. Variables with such p-values are considered statistically significant and valuable to the model.

R-squared (r^2) measures how well the regression model fits the data, indicating the proportion of variability in the dependent variable accounted for by the independent variables. An r^2 value of 0.942322855 means that roughly 94.23% of the variability in your dependent variable can be explained by the model.

Each variable's coefficient represents the estimated change in the dependent variable when the corresponding independent variable increases by one unit, assuming other variables remain unchanged.